Asssignment 6

Physics 451

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Question 1

Consider a rotor in two dimensions with the Hamilton operator

$$H = -\frac{\hbar^2}{2I}\frac{\mathrm{d}^2}{\mathrm{d}\theta^2} \tag{1.1}$$

and $0 \leq \theta \ll 2\pi$.

Question 1.a

Find the eigenstates and energy levels of the system

Here we need to solve:

$$H\psi = E\psi \tag{1.2}$$

Where ψ is the wavefunction and E the energy. This gives us:

$$-\frac{\hbar^2}{2I}\frac{\mathrm{d}^2\psi}{\mathrm{d}\theta^2} = E\psi \tag{1.3}$$

We then get a differential equation of the form:

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}\theta^2} = -\frac{2EI}{\hbar^2}\psi \tag{1.4}$$

We can then condense $2EI/\hbar^2 = m$. This differential equation has general solution:

$$\psi = Ae^{im\theta} + Be^{-im\theta} \tag{1.5}$$

To solve, we will take advantage of periodic boundary conditions where we need $\psi(\theta) = \psi(\theta + 2\pi)$. This gives us

$$Ae^{im\theta} + Be^{-im\theta} = Ae^{im(\theta+2\pi)} + Be^{-im(\theta+2\pi)} \tag{1.6}$$

We can then further solve:

$$Ae^{im\theta} - Ae^{im(\theta+2\pi)+Be^{-im\theta}-Be^{-im(\theta+2\pi)}} = 0 \tag{1.7}$$

$$Ae^{im\theta}(1 - e^{im2\pi}) + Be^{-im\theta}(1 - e^{-im2\pi}) = 0 \tag{1.8}$$

Where we need both terms to equal zero to obtain a solution, let's zoom in on this $e^{im2\pi}$ term.

$$e^{im2\pi} = \cos(m2\pi) + i\sin(m2\pi) \tag{1.9}$$

To make this equal to 1, (so that the whole factor goes to zero) we must have m be an integer so that the imaginary part goes to zero. Since m takes on both positive and negative values, we can drop the B term and have that:

$$\psi = Ae^{im\theta} \tag{1.10}$$

Now let's normalize to obtain A

$$1 = A^2\int_0^{2\pi} |\psi\psi^*|^2\,\mathrm{d}\theta = A^2\int_0^{2\pi}\left|\frac{e^{im\theta}}{e^{im\theta}}\right|^2\mathrm{d}\theta = A^2 2\pi \tag{1.11}$$

And so $A = 1/\sqrt{2\pi}$. We then get our eigenstates:

$$\psi = \frac{1}{\sqrt{2\pi}} e^{im\theta}, m = 0, \pm 1, \pm 2, \pm 3... \tag{1.12}$$

Where our energies are then solved with the original equation:

$$\frac{\mathrm{d}^2\psi}{\mathrm{d}\theta^2} = -\frac{2EI}{\hbar^2}\psi \tag{1.13}$$

$$-m^2\psi = -\frac{2EI}{\hbar^2}\psi \tag{1.14}$$

$$E = \frac{\hbar^2 m^2}{2I} \tag{1.15}$$

Where these are also quantized as m takes on integer values.

Question 1.b

Write the expression for the density matrix $\langle\theta'|\rho|\theta\rangle$ in a canonical ensemble of temperature T, and evaluate its low- and high-temperature limits. Discuss your results.

We have:

$$\langle\theta'|\rho|\theta\rangle = \frac{1}{2\pi Z}\sum_{m=-\infty}^{\infty} e^{-\beta\hbar^2 m^2/2I} e^{im(\theta'-\theta)} \tag{1.16}$$

With the partition function being:

$$Z = \sum_{m=-\infty}^{\infty} e^{-\beta\hbar^2 m^2/2I} \tag{1.17}$$

The low temperature limit ($T \to 0$) lets $\beta \to \infty$. This lets $Z = 1$ and a similar result for the top sum. So then:

$$\langle\theta'|\rho|\theta\rangle = \frac{1}{2\pi} \tag{1.18}$$

This flat result (no dependence on anything else) as well with what we generally know with systems of low energy is that this is the ground state. Now for the high temperature limit where ($T \to \infty, \beta \to 0$).

Question 2

Shining an intense laser beam on a semiconductor can create a metastable col- lection of electrons (charge $-e$, and effective m_e) and holes (charge $+e$, and effective mass m_h) in the bulk. The oppositely charged particles may pair up (as in a hydrogen atom) to form a gas of *excitons*, or they may dissociate into a plasma. We shall examine a much simpler model of this process

Question 2.a

Calculate the free energy of a gas composed of N_e electrons and N_h holes, at a temperature T, treating them classical non-interacting particles of masses m_e and m_h

The partition function for a given species with these restraints are:

$$Z = \frac{1}{N!}\left(\frac{V}{\lambda^3}\right)^N \tag{2.1}$$

Where $\lambda = \frac{h}{\sqrt{2\pi m k_B T}}$. We have then the free energy:

$$F = -k_B T \ln(Z) \tag{2.2}$$

Since we have two non interacting species, the total partition function is the product of the two (and thus the free energy is additive). Let's focus on the electrons.

$$F = -k_B T(\ln(Z_e) + \ln(Z_h)) \tag{2.3}$$

Where:

$$\ln(Z_e) = \ln\left(\frac{1}{N_e!}\left(V\left(\frac{m_e k_B T}{2\pi h^2}\right)^{3/2}\right)^{N_e}\right) \tag{2.4}$$

Using stirling's and expanding:

$$ln(Z_e) = N_e \ln\left(V\left(\frac{m_e k_B T}{2\pi h^2}\right)^{3/2}\right) - N_e \ln(N_e) + 1N_e \tag{2.5}$$

Which gives:

$$= N_e \ln\left(\frac{V}{N_e}\left(\frac{m_e k_B T}{2\pi h^2}\right)^{3/2}\right) + 1 \tag{2.6}$$

Finally:

$$F_e = -N_e k_B T\left(\ln\left(\frac{V}{N_e}\left(\frac{m_e k_B T}{2\pi h^2}\right)^{3/2}\right) + 1\right) \tag{2.7}$$

This is symmetric for the holes, so we then have:

$$F = F_e + F_h = -N_e k_B T\left(\ln\left(\frac{V}{N_e}\left(\frac{m_e k_B T}{2\pi h^2}\right)^{3/2}\right) + 1\right) - N_h k_B T\left(\ln\left(\frac{V}{N_h}\left(\frac{m_e k_B T}{2\pi h^2}\right)^{3/2}\right) + 1\right) \tag{2.8}$$

Question 2.b

By pairing into an exciton, the electron hole pair lowers its energy by ϵ. (The binding energy of a hydrogen-like exciton is $\epsilon \approx me^4/(2\hbar^2\epsilon^2)$, where ϵ is the dielectric constant, and $m^{-1} = m_e^{-1} + m_h^{-1}$.)Calculate the free energy of a gas of N_p excitons, treating them as classical non-interacting particles of mass $m = m_e + m_h$.

The slight modification comes in the form of:

$$Z_p = \frac{1}{N_p!}\left(\frac{V}{\lambda^3}e^{\epsilon\beta}\right)^{N_p} \tag{2.9}$$

Where the natural log is then:

$$\ln(Z_p) = N_p \ln\left(\frac{V}{\lambda^3} e^{\epsilon\beta}\right) - N_p \ln(N_p) + N_P \tag{2.10}$$

This then becomes:

$$\ln(Z_p) = N_p(\ln\left(\frac{V}{N_p\lambda^3}\right) + 1 + \epsilon\beta) \tag{2.11}$$

And so the free energy:

$$F_p = -N_p k_B T\left(\ln\left(\frac{V}{N_p\lambda^3}\right) + 1 + \epsilon\beta\right) \tag{2.12}$$

Where our λ has the sum of the electron and hole mass, instead of a single mass.

Question 2.c

Calculate the chemical potentials μ_e, μ_h, and μ_p of electron, hole, and exciton states, respectively.

The chemical potentials are just the following:

$$\mu = \frac{\partial F}{\partial N} \tag{2.13}$$

So applying this to the electron:

$$\mu_e = \frac{\partial F_e}{\partial N_e} \tag{2.14}$$

Ugh this is going to be gross.

$$\begin{aligned}
\frac{\partial F_e}{\partial N_e} &= -k_B T \frac{\partial}{\partial N_e} N_e \left(\ln\left(\frac{V}{N_e}\left(\frac{m_e k_B T}{2\pi h^2}\right)^{3/2}\right) + 1\right) && (2.15)\\
&= -k_B T \frac{\partial}{\partial N_e}\left(N_e \ln\left(\frac{V}{N_e}\left(\frac{1}{\lambda_e^3}\right)\right) + N_e\right) && (2.16)\\
&= -k_B T \frac{\partial}{\partial N_e}\left(N_e \ln\left(V\left(\frac{1}{\lambda_e^3}\right)\right) - N_e \ln(N_e) + N_e\right) && (2.17)\\
&= -k_B T(\ln\left(V\left(\frac{1}{\lambda_e^3}\right)\right) - \ln(N_e) - 1 + 1) && (2.18)\\
&= -k_B T \ln\left(\frac{V}{N_e}\left(\frac{1}{\lambda_e^3}\right)\right) && (2.19)
\end{aligned}$$

Because the hole free energy have a similar shape to the electron free energy, we get an equivalent chemical potential.

$$\mu_h = -k_B T \ln\left(\frac{V}{N_h}\left(\frac{1}{\lambda_h^3}\right)\right) \tag{2.20}$$

For μ_p we will derive it:

$$\mu_p = \frac{\partial F_p}{\partial N_p} \tag{2.21}$$

$$= -\frac{\partial}{\partial N_P} N_p k_B T \left(\ln\left(\frac{V}{N_p \lambda^3}\right) + 1 + \epsilon\beta \right) \tag{2.22}$$

$$= -k_B T \frac{\partial}{\partial N_P} \left(N_p \left(\ln\left(\frac{V}{\lambda^3}\right) - \ln(N_p) + 1 + \epsilon\beta \right) \right) \tag{2.23}$$

$$= -k_B T \frac{\partial}{\partial N_P} \left(N_p \ln\left(\frac{V}{\lambda^3}\right) - N_p \ln(N_p) + N_p + N_p \epsilon\beta \right) \tag{2.24}$$

$$= -k_B T \left(\ln\left(\frac{V}{N_p \lambda^3}\right) + \epsilon\beta \right) \tag{2.25}$$

Question 2.d

Express the equilibrium condition between excitons and electron/holes in terms of their chemical potentials.

We need the exchange of the electrons and holes to be equal to the exciton exchange. This means that:

$$\mu_e + \mu_h = \mu_p \tag{2.26}$$

References